AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM352
Web, Mobile and Cloud Technologies

Final Examination
Spring Semester 2025/2026

Date: 11/06/2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed: 3 Hrs**

Instructions:

- Total Marks: **100**
- This exam consists of **03** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART I (20 marks)

Answer all the questions
Choose the most correct option – 2 marks each

1 In virtualization, what does a virtual machine (VM) represent?
A. A physical computer
B. A simulated computer running on a host system
C. A network router
D. A database server

2. Which component is responsible for distributing incoming requests among multiple servers?
A. DNS server
B. Load balancer
C. Firewall
D. Hypervisor

3. What is the purpose of Network Time Protocol (NTP) in traditional infrastructure?
A. Encrypt network data
B. Synchronize clocks of servers
C. Store database records
D. Manage IP addresses

4. In AWS architecture, which service distributes traffic across servers and availability zones?
A. CloudFront
B. Elastic Load Balancing (ELB)
C. S3
D. Route 53

5. What is the main advantage of containers compared to virtual machines?
A. They require a separate operating system
B. They are larger in size
C. They do not require a separate operating system and are lightweight
D. They cannot run applications

6. Which of the following is NOT one of the three principles of responsive web design?
A. Fluid design
B. Flexible images
C. CSS media queries
D. Fixed-width layout

7. Which of the following is the recommended approach for designing a website that works well on both desktop and mobile devices?
A. Do nothing and rely on users to zoom
B. Maintain separate websites for desktop and mobile
C. Design a single responsive website that adjusts to all screen sizes
D. Use only fixed-width designs

8. Which of the following is NOT one of the three main principles of responsive web design?
A) Fluid design
B) Flexible images
C) CSS media queries
D) Server-side browser detection

9. Which of the following authentication is based on or using MD5 hash function
A. Basic authentication
B. Digest authentication
C. Client certificates.
D. None of the other given options.

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10. _____ is when data is protected so that it is infeasible for its contents to be changed in transit without any such changes being instantly obvious to the recipient.

A. Privacy
C. Integrity
B. Authenticity
D. Authorization

PART II (30 marks)

Answer the following questions

1. Describe the key components of a traditional infrastructure design for a web application. In your answer, explain the role of NTP, DNS, Router/NAT, Firewall/DMZ, Load Balancer, and Database server. [10 Marks]
2. Compare the OpenStack and Amazon Web Services (AWS) cloud architectures. In your answer, discuss the key components of each platform, how they provide compute, networking, storage, DNS, load balancing. [10 Marks]
3. Discuss the main classes of security threats that can affect web services in a distributed architecture. For each class, provide a clear definition and a practical real-world example. [8 marks]
4. Describe any two solutions from which mobile web designers can choose to address the screen size issue ages. [2 marks]

PART III (50 marks)

Question 1:
Read the scenario and solve the given tasks.

Scenario:

Read the scenario and solve the given four tasks.

A calorie calculator service is part of a personal health assistant system designed to help individuals monitor their daily nutrient intake. When a user enters the type and quantity of food they have consumed, the service calculates the total calories for those food items.

To perform this calculation, the system reads data from a file called Calories.json, which stores a list of different food items along with the number of calories in a standard serving size. By using this information, the service estimates the total calorie intake based on the amount of food entered by the user, helping them keep track of their diet and maintain a healthy lifestyle.

For example the Calories.json file should look like that

```
“Calories": [
  {
    "Food": "Apple",
    "Serving Size": "1 (4 oz.)",
    "Calories": "59"    },
  {
    "Food": "Broccoli",
    "Serving Size": "1 cup",
    "Calories": "45"    },
  ……..……]
```

Calories.json

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Tasks:

1. Implement the service as RESTful service with the required annotations. The service should accept the food type and the amount of serving size eaten and should return the sum of calories gained as HTML. [20 Marks]
2. Design the HTML page and write the HTML code that would consume or call the service. [12 Marks]
3. Specify the parameter type you have used in your implementation and the HTTP method used to call the service. [4 Marks]
4. In case that the web application should be available to mobile users, compare the development of the application as web, native, or hybrid application regarding the use of mobile hardware capabilities. Suggest the hardware capabilities that could be used for this application. [4 Marks]
5. Convert the following data give in the table form into JSON format. [10 Marks]

ID	Type	name	Image			Thumbnail		
			Image. width	Image.height	Image. URL	Thumbnail .width	Thumbnail. height	thumbnail.url
1	Donut	Cake	200	200	images/ 0001.jpg	32	32	images/thumbnails/0001.jpg
2	Pasta	chawmin	230	230	images/ 0002.jpg	35	35	images/thumbnails/0002.jpg

Table 1. product data

End of questions

Some Json class methods that could be used:

- static JsonArrayBuilder createArrayBuilder()//Creates a JSON array builder
- static JsonObjectBuilder createObjectBuilder()//Creates a JSON object builder
- static JsonReader createReader(InputStream in)//Creates a JSON reader from a byte stream.
- static JsonReader createReader(Reader reader) //Creates a JSON reader from a character stream.
- static JsonWriter createWriter(OutputStream out) //Creates a JSON writer to write a JSON object or array structure to the specified byte stream.
- static JsonWriter createWriter(Writer writer) //Creates a JSON writer to write a JSON object or array structure to the specified character stream.

Some JsonObject methods that could be used:

- boolean getBoolean(String name) //Returns the boolean value of the associated mapping for the specified name.

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- int getInt(String name)//returns int value to which the specified name is mapped.
- JsonArray getJsonArray(String name)//Returns the array value to which the specified name is mapped.
- JsonObject getJsonObject(String name)//Returns the object value to which the specified name is mapped.
- String getString(String name)//Returns the string value to which the specified name is mapped

Some JsonArray methods that could be used

- int getInt(int index) //A convenience method for getJsonNumber(index).intValue().
- JsonArray getJsonArray(int index) //Returns the array value at the specified position in this array.
- JsonObject getJsonObject(int index) //Returns the object value at the specified position in this array.
- String getString(int index) //Returns the String value at the specified position in this array.

Some JsonReader methods that could be used

- void close() //Closes this reader and frees any resources associated with the reader.
- JsonArray readArray() //Returns a JSON array that is represented in the input source.
- JsonObject readObject() //Returns a JSON object that is represented in the input source.

Some JsonWriter methods that could be used

- void close() //Closes this JSON writer and frees any resources associated with the writer.
- void writeArray(JsonArray array) //Writes the specified JSON array to the output source.
- void writeObject(JsonObject object) //Writes the specified JSON object to the output source.

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